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Mysunat Islam

Biomedical Engineering Undergraduate · MIST · Bioinformatics & Biomedical AI

Remote Biomedical Software Engineer · Biosignal Processing Engineer · Digital Health IoT Developer · Bioinformatics Data Analyst

Summary

Biomedical Engineering undergraduate at MIST specializing in turning complex biological and physiological data — ECG, EEG, motion signals, sensor streams, and biological datasets — into practical engineering solutions. Builds complete biomedical pipelines spanning sensor-level data acquisition, embedded hardware, signal processing, AI analysis, and interactive web-based visualization.

Strong technical foundation with leadership roles across multiple university clubs and proven success in competitive engineering and case-study projects. Comfortable working independently across hardware, signal processing, AI, and visualization layers without requiring separate specialist teams.

Core Technical Strengths

01 Biomedical Signal Processing

ECG/EEG/EMG acquisition, noise removal, baseline wander correction, R-peak detection, HRV feature extraction, CWT-based time-frequency analysis.

MATLAB · Python

02 Healthcare IoT & Embedded Hardware

End-to-end wearable rehabilitation and monitoring devices — FSR/IMU sensing, sensor fusion, servo & pneumatic actuation, wireless data streaming, and real-time browser-based digital twin visualization.

ESP32 · Arduino · BLE · JavaScript · Three.js · HTML/CSS

03 Medical Imaging & AI

U-Net segmentation, CNN classification, image normalization & morphological preprocessing for digital pathology and microscopy analysis.

Python · PyTorch · OpenCV · MATLAB

Skills

PROGRAMMING

Python · MATLAB · R · C/C++

AI / ML

PyTorch · TensorFlow · OpenCV · Deep Learning

BIOINFORMATICS

Bioconductor · Cytoscape

COMPUTATIONAL DRUG DESIGN

Molecular Docking · PyMOL · PyRx · Avogadro · Schrödinger

EMBEDDED / IOT

ESP32 · Arduino · BLE

WEB / VISUALIZATION

HTML · CSS · JavaScript · Three.js

ENGINEERING / SIMULATION

SolidWorks · Ansys · COMSOL Multiphysics · Simulink · Simscape Multibody

DESIGN / DOCS

Figma · Canva · LaTeX

04 Bioinformatics & Computational Biology

Gene expression analysis, differential expression workflows, PPI network mapping, molecular docking, and structure-based drug design.

R · Bioconductor · Cytoscape · PyMOL · PyRx · Avogadro · Schrödinger

05 Digital Twin & Biomedical Simulation

Closed-loop biomechanical models, neuromuscular control simulation, and human-machine interaction validation before hardware deployment.

MATLAB Simulink · Simscape Multibody

Experience

Executive — MIST Career Club
2024 – PRESENT

- Organized and managed key events and competitions.
- Campus Ambassador (CA) at Biztigation and organizer at FineAnswers.
- Coordinated logistics, speaker engagement, and participant communication for large-scale events.

Supervisor — MIST Robotics Club
2024 – PRESENT

Junior Executive — Hult Prize at MIST
2024 – PRESENT

- Contributed to the campus round of the international Hult Prize competition.
- Supported teams in business idea development, event coordination, and marketing efforts.

Executive — MIST Einthoven Club
2025

Interests

Bioinformatics · Signal Processing · Biodata Analysis · Digital Health

Education

BSc in Biomedical Engineering
MIST

2023 – 2027 · CGPA 3.66/4.00 (6 semesters completed)

Focus: Biomedical Signal Processing, Bioinstrumentation, Computational Biology

Higher Secondary Certificate (HSC)
Holy Cross College, Dhaka

2020 – 2022 · Science · GPA 5.00

Organizing member, HCC Science Fest 2022

Secondary School Certificate (SSC)
Ideal School and College

2020 · Science · GPA 5.00

Certifications

Bioinformatics for Drug Design & Development
Online course certificate

R for Bioinformatics & Transcriptomic Data Analysis
Online course certificate

Digital copy pending

EngiBiz Finalist — SheSTEM
National business case competition, MIST Career Club

Selected Projects

NeuroMap BD — Intent-Based Rehabilitation System with Real-Time Digital Twin

HEALTHCARE IOT | WEB VISUALIZATION

Full hardware-to-software rehabilitation pipeline: wearable hand glove + ankle sensing module with FSR pressure sensing, IMU motion tracking, sensor-fusion intent detection, and servo-assisted finger actuation. Built a real-time browser-based 3D digital twin (JavaScript, Three.js, CSS) to visualize patient motion live.

Digital twin video → Web app video → Hardware →

Closed-Loop Gait Monitoring, Pneumatic Trunk Support & Vibration System

HEALTHCARE IOT | DIGITAL TWIN

Distributed embedded architecture with ESP32 wireless sensor nodes, MPU6050 inertial tracking, and FSR plantar pressure measurement, driving PWM-controlled pneumatic and vibration feedback in a closed loop (motion detection → processing → abnormality ID → assistive feedback). Physical response loop validated in Simscape Multibody.

System digital twin →

Omics Biomarker Discovery & Biological Network Analysis Pipeline

BIOINFORMATICS

R-based pipeline for gene expression normalization, differential expression analysis, and PPI network mapping in Cytoscape, applied to early biomarker detection in pancreatic ductal adenocarcinoma.

View write-up →

AI-Based Biomedical Image Analysis Pipeline

IMAGING

PyTorch / U-Net segmentation pipeline with normalization and morphological preprocessing for automated cellular and tissue-level pattern recognition in fluorescence microscopy images.

View write-up →

ECG-Based Physiological State Analysis & Biomarker Extraction

SIGNAL PROCESSING

MATLAB pipeline with Butterworth bandpass & notch filtering, Pan-Tompkins R-peak detection, RR-interval analysis, and SDNN/RMSSD HRV feature extraction from single-lead ECG.

What I Deliver

- ★ End-to-end biomedical signal processing pipelines (ECG, EEG, EMG)
- ★ Wearable healthcare IoT devices with real-time web-based digital twins
- ★ Machine learning models for classification & prediction in health datasets
- ★ MATLAB / Python analysis, simulation & visualization tools
- ★ Reproducible, research-to-product workflows with clear documentation

References

Prof. Dr. Kaiissar Mannoor
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Normal state video →

Neuromechanical Digital Twin & Closed-Loop Simulation Platform

DIGITAL TWIN

Simulink / Simscape Multibody closed-loop biomechanical model with neuromuscular control and trajectory tracking, validating robotic hand interaction strategies before hardware testing.

NeuroRoboTwin →

Biomedical Device Optimization & Statistical Validation Framework

STATISTICS

DoE-based ANOVA and Tukey post-hoc analysis evaluating nanoparticle and carrier efficiency for drug delivery transport and controlled-release performance.

View presentation →

CAD Modeling — Artificial Extracorporeal Lung Assist Device

DESIGN

SolidWorks component design (valves, chambers, casing) for an artificial extracorporeal lung assist device, focused on biomedical design constraints and functional accuracy.

Analog ECG Acquisition Circuit Design

HARDWARE

AD620-based analog ECG acquisition circuit with 72 Hz high-pass and 796 Hz low-pass filter stages — Bioinstrumentation course project.

Open to remote collaboration with digital health startups, biomedical research teams, healthcare AI companies, medical device developers, and bioinformatics projects.